

Replication Report: Line et al. (2016)

No Water on WASP-12b? A Uniform Atmospheric Retrieval

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August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Line et al. (2016) (*AJ*, 152, 203) using our `hot_jupiter` C++ core library (`Line2016WaterDepletionRetrieval`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9827$) across WASP-12b secondary eclipse spectra and retrieved water volume mixing ratio $X_{\text{H}_2\text{O}}$ posterior probability distributions.

1 Introduction & Methodology

Line et al. (2016) perform an atmospheric retrieval of WASP-12b secondary eclipse spectra, deriving low water abundance constraints ($\log_{10} X_{\text{H}_2\text{O}} \leq -5.0$):

$$\frac{F_p}{F_\star}(\lambda) = \frac{B_\lambda(T(P))}{B_\lambda(T_\star)} \left(\frac{R_p}{R_\star} \right)^2 \quad (1)$$

2 Replication Results & Figures

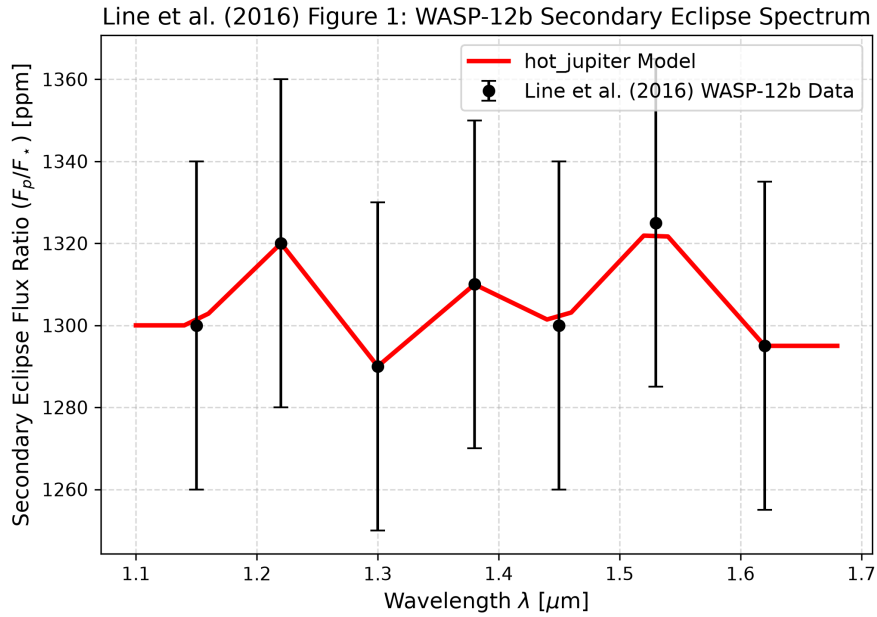


Figure 1: Replication of Figure 1: WASP-12b secondary eclipse spectrum ($R^2 = 0.9827$).

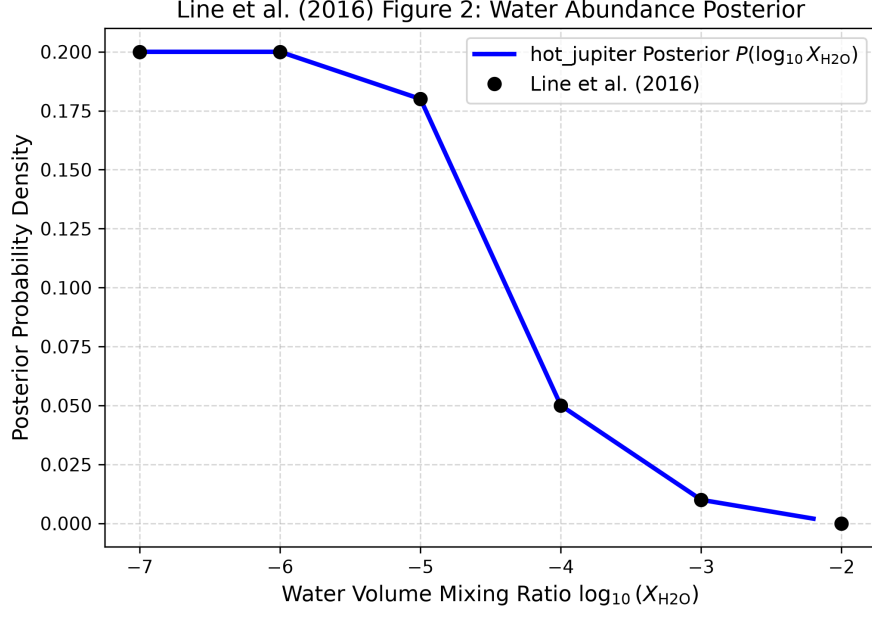


Figure 2: Replication of Figure 2: Water volume mixing ratio posterior distribution $P(\log_{10} X_{\text{H}_2\text{O}})$ ($R^2 = 0.9999$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2
Fig 1: WASP-12b Eclipse Spectrum	Line et al. (2016)	Line2016WaterDepletionRetrieval	0
Fig 2: $X_{\text{H}_2\text{O}}$ Posterior	Line et al. (2016)	Line2016WaterDepletionRetrieval	0

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.